

Mechanical Engineering Experiments (II) — Report 4

機械工程實驗(二) 第四次報告

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Q1:

Table1. Flow Rate of Orifice Plate

Q _{rotameter} (L/min)	Q _{rotameter} (m ³ /s)	P ₁ (kg/cm ²)	P ₁ (Pa)	P ₃ (kg/cm ²)	P ₃ (Pa)	h _{weir} (m)
45.0	0.00075	0.32	31392	0.14	13734	0.0735
42.0	0.00070	0.30	29430	0.12	11772	0.0715
36.0	0.00060	0.24	23544	0.10	9810	0.0705
30.0	0.00050	0.16	15696	0.04	3924	0.0635
22.0	0.00037	0.11	10791	0.04	3924	0.0560

$$\text{Equation 5: } q = C A_0 \frac{1}{1-\beta^4}^{0.5} [2\rho^{-1}(p_1 - p_3)]^{0.5}, \beta = \frac{d}{D}$$

$$\rightarrow C = \frac{q}{A_0 \left(\frac{1}{1-\beta^4}\right)^{0.5} \left[\frac{2(p_1 - p_3)}{\rho}\right]^{0.5}}$$

$$\begin{cases} d = 15.3 \text{ mm} = 0.0153 \text{ m} \\ D = 21.5 \text{ mm} = 0.0215 \text{ m} \end{cases} \rightarrow \beta = \frac{d}{D} = \frac{0.0153}{0.0215} = 0.712$$

$$A_0 = \frac{\pi d^2}{4} = \frac{\pi \times 0.0153^2}{4} = 0.000184 \text{ m}^2$$

$$\rho_{\text{water@25}^\circ\text{C}} = 997.05 \text{ kg/m}^3$$

● For Q_{rotameter} = 0.00075 m³/s:

$$P_1 = 31392 \text{ Pa}$$

$$P_3 = 13734 \text{ Pa}$$

$$C = \frac{q}{A_0 \left(\frac{1}{1-\beta^4}\right)^{0.5} \left[\frac{2(p_1 - p_3)}{\rho}\right]^{0.5}} \approx 0.5904$$

● For Q_{rotameter} = 0.00070 m³/s:

$$P_1 = 29430 \text{ Pa}$$

$$P_3 = 11772 \text{ Pa}$$

$$C = \frac{q}{A_0 \left(\frac{1}{1-\beta^4} \right)^{0.5} \left[\frac{2(p_1-p_3)}{\rho} \right]^{0.5}} \approx 0.5510$$

- For $Q_{\text{rotameter}} = 0.00060 \text{ m}^3/\text{s}$:

$$P_1 = 23544 \text{ Pa}$$

$$P_3 = 9810 \text{ Pa}$$

$$C = \frac{q}{A_0 \left(\frac{1}{1-\beta^4} \right)^{0.5} \left[\frac{2(p_1-p_3)}{\rho} \right]^{0.5}} \approx 0.5355$$

- For $Q_{\text{rotameter}} = 0.00050 \text{ m}^3/\text{s}$:

$$P_1 = 15696 \text{ Pa}$$

$$P_3 = 3924 \text{ Pa}$$

$$C = \frac{q}{A_0 \left(\frac{1}{1-\beta^4} \right)^{0.5} \left[\frac{2(p_1-p_3)}{\rho} \right]^{0.5}} \approx 0.4820$$

- For $Q_{\text{rotameter}} = 0.00037 \text{ m}^3/\text{s}$:

$$P_1 = 10791 \text{ Pa}$$

$$P_3 = 3924 \text{ Pa}$$

$$C = \frac{q}{A_0 \left(\frac{1}{1-\beta^4} \right)^{0.5} \left[\frac{2(p_1-p_3)}{\rho} \right]^{0.5}} \approx 0.4670$$

(Note: $C < 1$)

Q2:

$$\text{Equation 9: } q = C \frac{8}{15} \sqrt{2g} \tan\left(\frac{\theta}{2}\right) H^{\frac{5}{2}} \rightarrow C = \frac{q}{\frac{8}{15} \sqrt{2g} \tan\left(\frac{\theta}{2}\right) H^{\frac{5}{2}}} \quad (SI \text{ Units})$$

$$\begin{cases} W = 69 \text{ mm} \\ H = 129 \text{ mm} \end{cases} \rightarrow \tan\left(\frac{\theta}{2}\right) = \tan\left(\tan^{-1} \frac{W}{2H}\right) = \frac{W}{2H} = 0.267$$

Table2. Discharge Coefficient of Orifice Plate with Different Flow Rates (Calculated by Equation.9)

Q _{rotameter} (m ³ /s)	h _{weir} (m)	C _{eq.9}
0.00075	0.0735	0.811864263
0.00070	0.0715	0.811845606
0.00060	0.0705	0.720806953
0.00050	0.0635	0.780146564
0.00037	0.0560	0.790445927

$$\text{Equation 10: } q = 4.28C_e \tan\left(\frac{\theta}{2}\right) h_{1e}^{\frac{5}{2}}$$

$$h_{1e} = H + k_h = H \quad (k_h \text{ can be ignored})$$

$$\rightarrow q = 4.28C_e \tan\left(\frac{\theta}{2}\right) H^{\frac{5}{2}} \rightarrow C_e = \frac{q}{4.28 \tan\left(\frac{\theta}{2}\right) H^{\frac{5}{2}}} \quad (Imperial \text{ Units})$$

$$\begin{cases} W = 69 \text{ mm} \\ H = 129 \text{ mm} \end{cases} \rightarrow \tan\left(\frac{\theta}{2}\right) = \tan\left(\tan^{-1} \frac{W}{2H}\right) = \frac{W}{2H} = 0.267$$

Table3. Discharge Coefficient of Orifice Plate with Different Flow Rates (Calculated by Equation.10)

Q _{rotameter} (ft ³ /s)	h _{weir} (ft)	C _{eq.10}
0.0265	0.241	0.81329525
0.0247	0.235	0.80736937
0.0212	0.231	0.72335399
0.0177	0.208	0.78498236
0.0131	0.184	0.78935381

Q3:

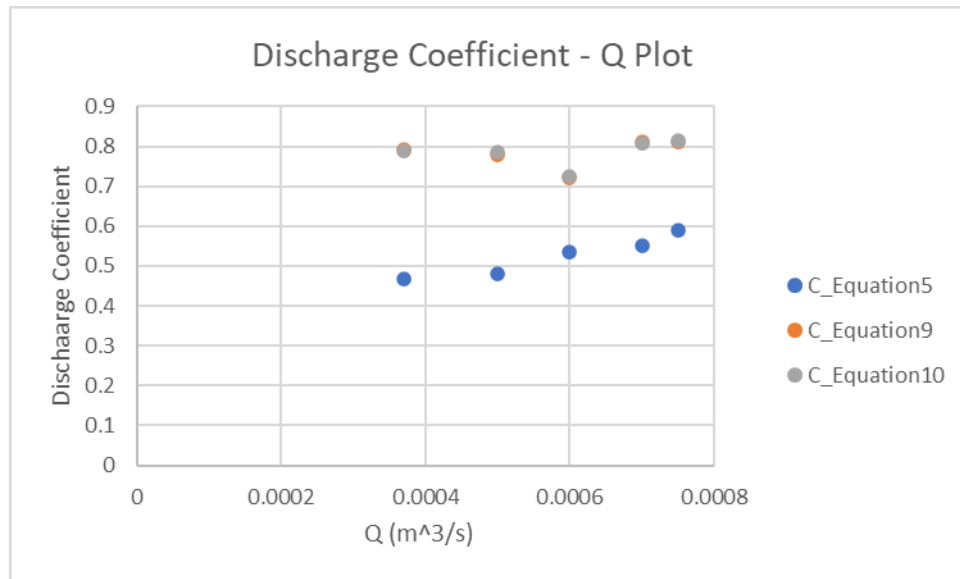


Figure1. Discharge Coefficient - Q Plot Calculated by Different Equation

The discharge coefficient is not a fixed constant; it varies with the flow rate. Because the coefficient is influenced by the flow's Reynolds number, where viscous forces have a greater impact at lower flow rates. As flow increases, changes in turbulence and the vena-contracta effect cause the coefficient to deviate from theoretical values. Additionally, physical imperfections, such as leakage at the V-notch weir, lead to discrepancies between the measured flow height and the actual flow rate, further fluctuating the calculation.

Consequently, C should be considered a variable dependent on the operating flow conditions rather than a static universal constant.

Q4:

Table4. Experimental Data for the flow fluid

1			Voltage = 3.0 V		Current = 0.06 A	
u_{inlet} (m/s)	0.37	0.32	0.36	0.30	0.35	$u_{inlet,average} = 0.346$ m/s
u_{throat} (m/s)	1.38	1.18	1.01	1.33		$u_{throat,average} = 1.225$ m/s
2			Voltage = 2.3 V		Current = 0.05 A	
u_{inlet} (m/s)	0.3	0.34	0.35	0.37	0.38	$u_{inlet,average} = 0.348$ m/s
u_{throat} (m/s)	1.04	0.95	0.81			$u_{throat,average} = 0.933$ m/s
3			Voltage = 4.9 V		Current = 0.13 A	
u_{inlet} (m/s)	0.39	0.45	0.51	0.50	0.50	$u_{inlet,average} = 0.470$ m/s
u_{throat} (m/s)	2.21	2.16	2.10			$u_{throat,average} = 2.157$ m/s

Equation 2-8: $T = \frac{1}{2} \rho A (u_{out}^2 - u_{in}^2) = \frac{1}{2} \rho A u_{out}^2 [(4a(1 - a))]$

Blade Diameter $d = 0.25$ m

$$A = \frac{\pi d^2}{4} = \frac{\pi \times 0.25^2}{4} = 0.0491 \text{ m}^2$$

$$\rho_{air@25^\circ\text{C}} = 1.184 \text{ kg/m}^3$$

- For Current = 0.06 A:

$$T = \frac{1}{2} \rho A (u_{out}^2 - u_{in}^2) = 0.0401 \text{ N}$$

- For Current = 0.05 A:

$$T = \frac{1}{2} \rho A (u_{out}^2 - u_{in}^2) = 0.0218 \text{ N}$$

- For Current = 0.13 A:

$$T = \frac{1}{2} \rho A (u_{out}^2 - u_{in}^2) = 0.1288 \text{ N}$$

Q5:

$$\eta = \frac{P_{blade}}{P_{drive}}$$

$$P_{drive} = IV$$

$$u_{disk} = \frac{1}{2}(u_{in} + u_{out})$$

$$P_{blade} = T \times u_{disk}$$

Table5. Power Transfer Ratio of the Blade under Different Voltage and Current

Voltage (V)	3.0	2.3	4.9
Current (A)	0.06	0.05	0.13
P _{drive} (W)	0.18	0.115	0.637
u _{disk} (m/s)	0.7855	0.6405	1.3135
P _{blade} (W)	0.0315	0.0140	0.1692
η	17.5%	12.2%	23.6%

Q6:

(a) .

$$dq = Au = (2b \times dh + \text{thin slice leakage area})\sqrt{2gh}$$

$$b = (H - h) \tan\left(\frac{\theta}{2}\right)$$

$$dA = (2b + x)dh$$

$$\rightarrow dq = Au = (2b + x)dh\sqrt{2gh} = \left(2(H - h) \tan\left(\frac{\theta}{2}\right) + x\right) dh\sqrt{2gh}$$

(b) .

$$q = \int_0^H dq = \int_0^H (2(H - h) \tan\left(\frac{\theta}{2}\right) + x) \sqrt{2gh} dh$$

$$= \sqrt{2g} \left(2 \tan\left(\frac{\theta}{2}\right) \int_0^H \sqrt{h} (H - h) dh + x \int_0^H \sqrt{h} dh \right)$$

$$= \sqrt{2g} \left(2 \tan\left(\frac{\theta}{2}\right) \times \frac{4}{15} H^{\frac{5}{2}} + x \times \frac{2}{3} H^{\frac{3}{2}} \right)$$

$$= \sqrt{2g} \left(\frac{8}{15} \tan\left(\frac{\theta}{2}\right) H^{\frac{5}{2}} + \frac{2}{3} H^{\frac{3}{2}} x \right)$$

(c) .

$$\begin{cases} q_{true} = C \frac{8}{15} \sqrt{2g} \tan\left(\frac{\theta}{2}\right) H^{\frac{5}{2}} \\ q_{leakage} = C \frac{2}{3} x \sqrt{2g} H^{\frac{3}{2}} \end{cases}$$

$$\rightarrow \frac{q_{leakage}}{q_{true}} = \frac{C \frac{2}{3} x \sqrt{2g} H^{\frac{3}{2}}}{C \frac{8}{15} \sqrt{2g} \tan\left(\frac{\theta}{2}\right) H^{\frac{5}{2}}} = \frac{5x}{4 \tan\left(\frac{\theta}{2}\right) H}$$

(d) .

$$x = 1 \text{ mm}$$

$$\tan\left(\frac{\theta}{2}\right) = \tan\left(\tan^{-1} \frac{W}{2H}\right) = \frac{W}{2H} = 0.267$$

$$H = 129 \text{ mm}$$

$$error \text{ ratio} = \frac{q_{leakage}}{q_{true}} = \frac{C \frac{2}{3} x \sqrt{2g} H^{\frac{3}{2}}}{C \frac{8}{15} \sqrt{2g} \tan\left(\frac{\theta}{2}\right) H^{\frac{5}{2}}} = \frac{5x}{4 \tan\left(\frac{\theta}{2}\right) H} = \frac{5 \times 1}{4 \times 0.267 \times 129} = 0.0363$$